

Robust Bayesian Analysis with Information Geometry and Riemannian Optimization

Application to Nonlinear Dynamics

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Outline

Introduction

Bayesian inference

Information Geometry-based Robust Bayesian Analysis

Application to nonlinear dynamics

Conclusion and perspectives

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Introduction

- **Bayesian inference** provides a comprehensive probabilistic framework for quantifying epistemic uncertainties and updating them from new information, typically measured data
(Gelman et al., 2013; Kaipio and Somersalo, 2005)

Introduction

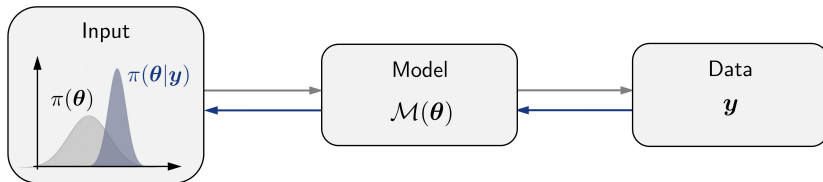
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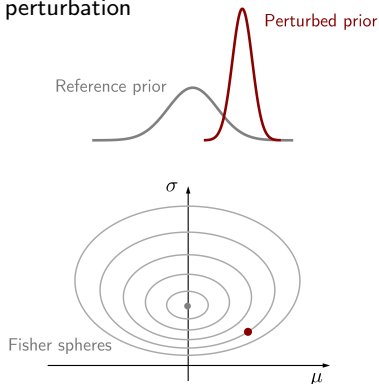
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- ▶ It is based on the definition of a **prior distribution**, which summarizes a state of knowledge about some input parameters before making any observation
- ▶ By conditioning on observed data, Bayesian inference enables to derive a **posterior distribution**, reflecting an updated state of knowledge about parameters
- ▶ In this framework, the selection of the prior distribution is of utmost importance !
⇒ Necessity to analyze the impact of the choice of the prior on Bayesian inference results
(Berger, 1990; Ruggeri et al., 2005)

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Main goal

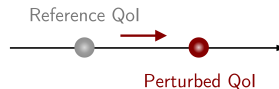
Definition of **Perturbed-law based sensitivity indices** dedicated to Bayesian inference, using the **Fisher distance** from information geometry to perturb prior distributions

Prior perturbation



Inference
 $\Rightarrow$

Posterior quantity of interest (QoI)



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General framework of Bayesian inference

- ▶ Let $\mathbf{X}, \mathbf{Y}$ be random variables respectively taking values in $\mathcal{X} \subset \mathbb{R}^d$ and $\mathcal{Y} \subset \mathbb{R}^n$, where:
 - $\mathbf{X}$ represents uncertain input **parameters** of a given model,
 - $\mathbf{Y}$ represents **observables**.
- ▶ Input parameters $\mathbf{X}$ are assumed to admit a density $\pi_{\mathbf{X}}$, known as the **prior** density
- ▶ Given observed data $\mathbf{y} \in \mathcal{Y}$, **Bayes' theorem** enables to derive the following expression for the **posterior** density (*i.e.*, the conditional density of $\mathbf{X}$ knowing $\mathbf{Y} = \mathbf{y}$):

$$\pi_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})}{C}$$

where:

- $\mathcal{L}(\mathbf{x}) = \pi_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x})$ is the **likelihood** (*i.e.*, the conditional density of $\mathbf{Y}$ knowing $\mathbf{X} = \mathbf{x}$, for $\mathbf{x} \in \mathcal{X}$),
- $C > 0$ is the **model evidence**:

$$C = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})d\mathbf{x}.$$

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Fisher distance between densities

- ▶ The prior distribution is assumed to admit independent marginals, and the associated densities $\pi_{X_1}, \dots, \pi_{X_d}$ are assumed to belong to the family of **parametric densities** $\mathcal{S} = \bigcup_{\Theta \subset \mathbb{R}^r, r \geq 1} \{\pi_{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \Theta\}$
- ▶ Given a **parametric space** $\Theta \subset \mathbb{R}^r$, with $r \geq 1$, a set of parameters $\boldsymbol{\theta} \in \Theta$ determines a unique density $\pi_{\boldsymbol{\theta}}$ in the manifold $\mathcal{S}_{\Theta} = \{\pi_{\tilde{\boldsymbol{\theta}}} \mid \tilde{\boldsymbol{\theta}} \in \Theta\}$
- ▶ $\mathcal{S}_{\Theta}$ defines a Riemannian manifold when equipped with the **Fisher metric** defined by: (Rao, 1945)

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}[(\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X})) (\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X}))^{\top}]$$

- ▶ The **Fisher distance** between two densities $\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2} \in \mathcal{S}_{\Theta}$ is then defined by:

$$d_F(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2}) = \min_{\gamma \in \mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})} \ell(\gamma)$$

where $\mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})$ is the set of paths joining $\pi_{\boldsymbol{\theta}_1}$ to $\pi_{\boldsymbol{\theta}_2}$, and $\ell(\gamma)$ the **length** (in Fisher metric sense) of γ

Fisher perturbations of densities

- ▶ A **geodesic** is a path which minimizes its length (in the Fisher metric sense) (Costa et al., 2015)

- ▶ Given $\pi_{\theta_0} \in \mathcal{S}_\Theta$ and $\theta_0 \in \Theta$, the **Fisher sphere** of center π_{θ_0} and radius $\delta > 0$ is defined by:

$$\partial\mathcal{B}_F(\pi_{\theta_0}, \delta) = \{\pi_\theta \in \mathcal{S}_\Theta \mid d_F(\pi_{\theta_0}, \pi_\theta) = \delta\}$$

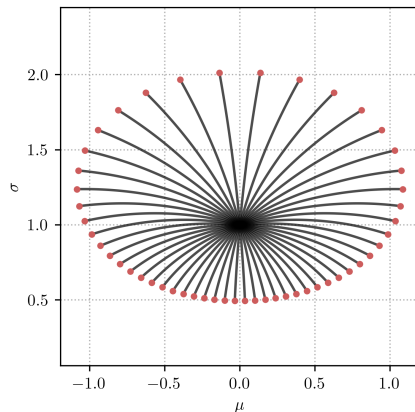
i.e., the collection of the endpoints of all geodesics starting from π_{θ_0} with length δ

- ⇒ **Definition of a perturbed density:** given a marginal π_{X_i} and a perturbation level $\delta > 0$, a perturbed density $\pi_{X_i, \delta}$ is defined by a point lying on the Fisher sphere $\partial\mathcal{B}_F(\pi_{X_i}, \delta)$ (Gauchy et al., 2022)

- ▶ Geodesics can be computed through solving a system of ordinary differential equations (ODE) known as **Hamilton's equations**, by using adapted **time integration** methods (Amari and Nagaoka, 2007; Deriglazov, 2010)

Fisher density perturbations - Example of Gaussian densities

- Example of the manifold of 1D Gaussian densities $\mathcal{S}_\Theta = \{\mathcal{N}(\cdot; \mu, \sigma^2) \mid (\mu, \sigma) \in \Theta\}$, with $\Theta = \mathbb{R} \times \mathbb{R}_{>0}$:



Samples on the Fisher sphere with center $(0, 1)$ and radius $\delta = 1$ (red dots), and associated geodesics (black lines)

Perturbed prior and posterior densities

- ▶ Definition of a perturbed prior joint density, given $i \in \{1, \dots, d\}$ and a perturbation level $\delta > 0$:

$$\pi_{i\delta}(\mathbf{x}) = \left(\prod_{1 \leq j \leq d, j \neq i} \pi_{X_j}(x_j) \right) \pi_{X_{i,\delta}}(x_i),$$

where $\pi_{X_{i,\delta}} \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)$ is a perturbation of the marginal density π_{X_i}

- ▶ The support of the perturbed prior is assumed to be a subset of the unperturbed prior:

$$\text{supp}(\pi_{i\delta}) \subseteq \text{supp}(\pi_{\mathbf{X}})$$

- ▶ The **perturbed posterior** density can then be written as follows:

$$\pi_{i\delta}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})}{C_{i\delta}}$$

where $C_{i\delta} > 0$ denotes the **perturbed model evidence**:

$$C_{i\delta} = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})d\mathbf{x}$$

Perturbed Law-based Indices for Bayesian inference

- Definition of **Bayesian Perturbed Law-based Indices** (BPLI), given a perturbation $\pi_{X_i, \delta} \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)$ with level $\delta > 0$ of the i -th prior marginal density: (Rossat, 2026)

$$S(\pi_{X_i, \delta}) = \frac{C_{i\delta} - C}{C}$$

- Definition of minimal and maximal BPLI over the Fisher sphere $\partial \mathcal{B}_F(\pi_{X_i}, \delta)$:

$$S_{i\delta}^- = \min_{\pi \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi) \qquad S_{i\delta}^+ = \max_{\pi \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi)$$

- ⇒ The range $S_{i\delta}^+ - S_{i\delta}^-$ enables to quantify the impact of a lack of knowledge in the prior of X_i on model evidence, for an increasing perturbation level δ

Practical computation of BPLI

- Based on the **BUS framework** (*Bayesian Updating with Structural reliability methods*), the BPLI $S(\pi_{X_i, \delta})$ can be rewritten as the relative variation of some failure probabilities (Straub and Papaioannou, 2015)

⇒ Enables the use of reverse importance sampling (IS) to estimate $S(\pi_{X_i, \delta})$, from N i.i.d. prior samples:

$$S(\pi_{X_i, \delta}) \approx \hat{S}_N(\pi_{X_i, \delta})$$

- Extremal BPLI $S_{i\delta}^-$ and $S_{i\delta}^+$ can then be estimated via two distinct methods:
 - **From Fisher sphere samples**, by drawing K samples $\{\pi_{X_i, \delta}^{(k)}\}_{1 \leq k \leq K} \subset \partial \mathcal{B}_F(\pi_{X_i, \delta})$ and then setting: (Rossat, 2026)

$$S_{i\delta}^- \approx \min_{1 \leq k \leq K} \hat{S}_N(\pi_{X_i, \delta}^{(k)}) \qquad S_{i\delta}^+ \approx \max_{1 \leq k \leq K} \hat{S}_N(\pi_{X_i, \delta}^{(k)})$$

- **With Riemannian optimization**, by directly estimating extremal values of $\hat{S}_N$ over $\partial \mathcal{B}_F(\pi_{X_i, \delta})$

Riemmanian optimization for extremal BPLI estimation

- Given a marginal index $i \in \{1, \dots, d\}$ and $\delta > 0$, one considers the functional $\hat{\mathcal{S}}_N : \Theta \rightarrow \mathbb{R}$ defined by:

$$\hat{\mathcal{S}}_N(\boldsymbol{\theta}) = \hat{S}_N(\pi_{\boldsymbol{\theta}})$$

for all $\boldsymbol{\theta} \in \Theta$, where $\pi_{\boldsymbol{\theta}} \in \partial\mathcal{B}_F(\pi_{\boldsymbol{\theta}_0}, \delta)$, and where $\boldsymbol{\theta}_0 \in \Theta$ are reference prior parameters (i.e., $\pi_{\boldsymbol{\theta}_0} = \pi_{X_i}$)

- Finding the extrema of $\hat{\mathcal{S}}_N$ over the Fisher sphere is akin to finding the extrema of the following functional defined on the unit sphere of $\mathbb{R}^r$:

$$\mathcal{J}(\mathbf{u}) = \hat{\mathcal{S}}_N(\exp_{\boldsymbol{\theta}_0}(\delta \mathbf{G}_{\boldsymbol{\theta}_0}^{1/2} \mathbf{u}))$$

for all $\mathbf{u} \in \mathbb{S}^{r-1} = \{\mathbf{v} \in \mathbb{R}^r \mid \mathbf{v}^\top \mathbf{v} = 1\}$ where:

- $\exp_{\boldsymbol{\theta}_0}$ is the **exponential mapping** (i.e., $\exp_{\boldsymbol{\theta}_0}(\mathbf{v})$ is the endpoint of the geodesic starting from $\boldsymbol{\theta}_0$ with velocity $\mathbf{v} \in \mathbb{R}^r$)
- $\mathbf{G}_{\boldsymbol{\theta}_0}^{1/2}$ is a square root of the inverse Fisher metric $\mathcal{I}(\boldsymbol{\theta}_0)^{-1}$

⇒ Estimation of extremal BPLI based on a (gradient-free) Riemannian CMA-ES optimization algorithm, through the computation of extrema of $\mathcal{J}$ on the sphere $\mathbb{S}^{r-1}$ (Colutto et al., 2010)

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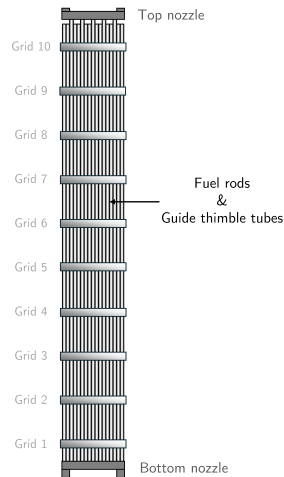
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Case of an impact test of a PWR fuel assembly

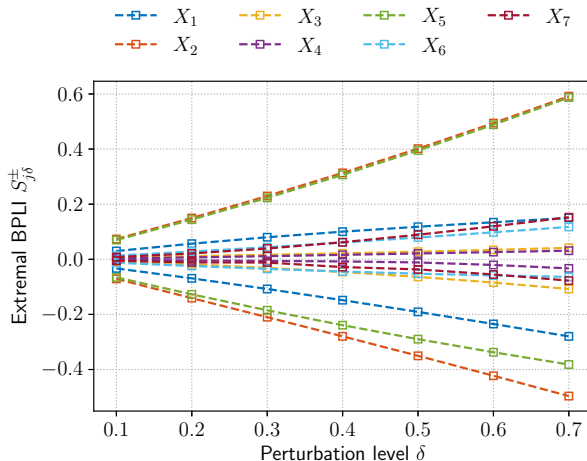
- ▶ Identification of parameters of a finite element model of a **pressurized water reactor** (PWR) fuel assembly, through an **impact test** (Yang et al., 2025)
- ▶ Imposed initial displacement at grids 4 and 7, and **single-point impact** at grid 5
- ▶ Finite element model based on beams and discrete elements representing the mechanical behavior of fuel rods, guide thimble tubes and grids
- ▶ Nonlinear transient calculations performed with **code_aster** solver (Électricité de France, 2026)
- ▶ Inference of **7 input parameters** describing stiffness and damping properties of the fuel assembly, from experimental data



View of the studied PWR fuel assembly

Robust Bayesian inference results

- Estimation of extremal BPLI, based on reverse IS with $N = 10^5$ prior samples and Riemannian CMA-ES:



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Conclusion

- Proposition of **Bayesian Perturbed Law-based Indices (BPLI)**, aiming at quantifying the impact of the choice of the prior distribution on results of Bayesian inference (Rossat, 2026)



Paper "*Information Geometry based Robust Bayesian Analysis*" available in IJUQ

Conclusion

- ▶ Proposition of **Bayesian Perturbed Law-based Indices (BPLI)**, aiming at quantifying the impact of the choice of the prior distribution on results of Bayesian inference (Rossat, 2026)
- ▶ BPLI can help to reduce the dimensionality of Bayesian inverse problems, by focusing on important input parameter
- ▶ BPLI estimation remains tractable to problems involving nonlinear forward models with possibly high-dimensional inputs, while allowing an arbitrary perturbation level for the prior
⇒ contrasts with a significant part of existing Robust Bayesian Analysis approaches (Betrò and Guglielmi, 2000)

Perspectives

- ▶ The likelihood ratio of reverse importance sampling can locally take high values, which can impede the quality of estimated BPLI and their confidence intervals (Gauchy et al., 2022)
- ▶ Similar problems in the case of small BUS failure probabilities (*i.e.* strongly informative data and/or high dimensional input distributions)
⇒ need for more efficient sampling techniques for estimating perturbed BUS failure probabilities (Au and Beck, 2001; Betz et al., 2018)
- ▶ BPLI estimation cost can be prohibitive in the case of costly forward models ⇒ use of **surrogate modeling** techniques (either *prior-based* or *adaptive*) to ensure a feasible computational cost (Marzouk and Najm, 2009; Wagner et al., 2021)

Thank you for your attention !

References I

- Amari, S. and Nagaoka, H. (2007). *Methods of Information Geometry*. American Mathematical Society.
- Au, S.-K. and Beck, J. L. (2001). Estimation of small failure probabilities in high dimensions by subset simulation. *Probabilistic Engineering Mechanics*, 16(4):263–277.
- Berger, J. O. (1990). Robust bayesian analysis: sensitivity to the prior. *Journal of Statistical Planning and Inference*, 25(3):303–328.
- Betrò, B. and Guglielmi, A. (2000). *Methods for Global Prior Robustness under Generalized Moment Conditions*, pages 273–293. Springer New York, New York, NY.
- Betz, W., Papaioannou, I., Beck, J. L., and Straub, D. (2018). Bayesian inference with subset simulation: Strategies and improvements. *Computer Methods in Applied Mechanics and Engineering*, 331:72–93.
- Colutto, S., Fruhauf, F., Fuchs, M., and Scherzer, O. (2010). The cma-es on riemannian manifolds to reconstruct shapes in 3-d voxel images. *IEEE Transactions on Evolutionary Computation*, 14(2):227–245.
- Costa, S. I., Santos, S. A., and Strapasson, J. E. (2015). Fisher information distance: A geometrical reading. *Discrete Applied Mathematics*, 197:59–69. Distance Geometry and Applications.
- Deriglazov, A. (2010). *Classical Mechanics*. Springer Berlin Heidelberg.

References II

- Gauchy, C., Stenger, J., Sueur, R., and Iooss, B. (2022). An information geometry approach to robustness analysis for the uncertainty quantification of computer codes. *Technometrics*, 64(1):80–91.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC.
- Kaipio, J. and Somersalo, E. (2005). *Statistical and Computational Inverse Problems*. Springer.
- Marzouk, Y. M. and Najm, H. N. (2009). Dimensionality reduction and polynomial chaos acceleration of bayesian inference in inverse problems. *Journal of Computational Physics*, 228(6):1862–1902.
- Rao, C. (1945). Information and the accuracy attainable in the estimation of statistical parameters. *Bull. Calcutta Math. Soc.*, 37.
- Rossat, D. (2026). Information-Geometry-Based Robust Bayesian Analysis. *International Journal for Uncertainty Quantification*, 16(2):47–73.
- Ruggeri, F., Ríos Insua, D., and Martín, J. (2005). Robust Bayesian Analysis. In Dey, D. and Rao, C., editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 623–667. Elsevier.
- Straub, D. and Papaioannou, I. (2015). Bayesian updating with structural reliability methods. *Journal of Engineering Mechanics*, 141(3):04014134.

References III

- Wagner, P.-R., Marelli, S., and Sudret, B. (2021). Bayesian model inversion using stochastic spectral embedding. *Journal of Computational Physics*, 436:110141.
- Yang, T., Zhang, Y., Cai, F., Zhang, K., and Lu, Z. (2025). Impact response analysis of fuel assembly based on nonlinear simplified dynamics modeling. *Progress in Nuclear Energy*, 180:105617.
- Électricité de France (1989–2026). `code_aster` - Analysis of structures and thermomechanics for studies and research. Open source on www.code-aster.org.